

HIREFLOW

Product Requirements Document

Project 2 · HR Tech · Version 3.0 · May 2026

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v3.0 — What's New

NEW	Interview Scoresheet with structured criteria (Technical, Behavioural, Communication, Culture Fit, Problem Solving)
NEW	AI match percentage per candidate shown during Screening
NEW	Job Posting status lifecycle (DRAFT, OPEN, PAUSED, CLOSED, FILLED)
NEW	Automated rejection emails include specific unmatched skills
NEW	Manual rejection requires Hiring Manager to input reason (mandatory)
NEW	All notification emails show current stage + next expected stage

5	5	4	5	6
PIPELINE STAGES	JOB LISTING STATUSES	SYSTEM ACTORS	SCORESHEET CRITERIA	CORE ENTITIES

01 Product Overview

HireFlow is an AI-augmented recruitment platform that enables companies to post job listings and manage candidates through a structured, multi-stage hiring pipeline. It combines AI-assisted resume screening (with match percentage scoring) with structured human decision-making, event-driven notifications via Kafka, and real-time admin analytics — reducing time-to-hire while keeping the process transparent and fully auditable.

Interviews are conducted externally on Google Meet. HireFlow handles slot creation, stores the Meet link, delivers it to the candidate, and collects a structured post-interview scoresheet from the Hiring Manager after the meeting.

02 Actors & Responsibilities

Actor	Role & Responsibilities
Applicant	Creates a profile, uploads a resume, and applies to open job listings. Receives stage-change notifications that always include current stage and next expected stage. Cannot apply to the same role twice.
Hiring Manager	Reviews AI-screened applications with match percentage, schedules interviews (InterviewSlot + Meet link), submits structured scoresheets post-interview, and records final decisions. Manual rejections require a mandatory reason.
Admin	Monitors platform via real-time dashboard — active listings, job posting status lifecycle, pipeline volume per stage, and time-to-hire metrics.
System	Executes AI screening with match percentage calculation, sends automated rejection emails with specific unmatched skills, enforces duplicate-application constraint, persists every ApplicationStageUpdate, and publishes Kafka events with current + next stage in every notification.

03 Core Data Entities

Entity	Description
User	Shared entity for all actors. Role-discriminated: APPLICANT HIRING_MANAGER ADMIN.
Company	Tenant entity. Each company owns its JobListings, hiring pipeline, and scoresheet templates.
JobListing	[UPDATED] Represents an open role. Now includes status field: DRAFT OPEN PAUSED CLOSED FILLED. Visible to Admin and Hiring Manager at all times.

Application	Links an Applicant to a JobListing. Holds current pipeline stage. Unique constraint on (applicant_id, job_listing_id).
ApplicationStageUpdate	Immutable audit log. Records every stage transition with actor, timestamp, reason, and next_stage for notification payloads.
InterviewSlot	[UPDATED] Scheduling record: date/time, meet_link (URL, required). Now includes full Scoresheet fields: technical_score, behavioural_score, communication_score, culture_fit_score, problem_solving_score, overall_recommendation, notes, decision (PASS REJECT), review_submitted_at.
AI ScreeningResult	[NEW] Stores AI output per application: match_percentage (0-100), matched_skills[], unmatched_skills[], summary_note. Linked to Application. Used for Hiring Manager view and automated rejection email content.

04 Job Listing Status Lifecycle [NEW]

Every JobListing has a status that reflects its current state in the recruitment cycle. Admins and Hiring Managers can see this status at all times on the listings dashboard.

Status	Description
DRAFT	Job listing created but not yet published. Hiring Manager is still editing the description and requirements.
OPEN	Listing is live and accepting applications. Applicants can find and apply to this role.
PAUSED	Listing temporarily hidden from applicants. No new applications accepted. Existing pipeline continues.
CLOSED	Listing no longer accepting applications. Pipeline in progress for existing applicants.
FILLED	Role has been filled. A candidate has been HIRED. Listing archived and removed from active view.

Status transitions: *DRAFT → OPEN → PAUSED → OPEN (re-open) | CLOSED | FILLED. FILLED is automatically set by System when an application reaches HIRED stage.*

05 Hiring Pipeline — Stage Flow

Every Application moves through these stages in order. Backwards transitions are not permitted. Each transition writes an ApplicationStageUpdate and fires a Kafka event. Every notification email includes both the current stage

and the next expected stage.

1	APPLIED Applicant submits profile. System parses resume, enforces duplicate guard, sends confirmation email with current stage (APPLIED) and next stage (SCREENING).	Actor Applicant / System Kafka Yes
2	SCREENING [UPDATED] AI scores resume and calculates match percentage (0-100%) vs job requirements. Matched and unmatched skills logged in AIScreeningResult. Clear passes advance automatically; borderline cases escalated to Hiring Manager. Auto-rejections include specific unmatched skills in the notification. Human rejections require mandatory reason input.	Actor System / Hiring Manager Kafka Yes
3	INTERVIEW SCHEDULED [UPDATED] Hiring Manager creates InterviewSlot with Google Meet link. System notifies applicant with slot details, Meet link, current stage, and next stage (OFFER_SENT or REJECTED). After Google Meet call, Hiring Manager returns to HireFlow to complete structured scoresheet.	Actor Hiring Manager / System Kafka Yes
4	OFFER SENT Offer extended following a positive scoresheet decision. Applicant notified with current stage (OFFER_SENT) and next stage (HIRED or REJECTED).	Actor Hiring Manager Kafka Yes
5	HIRED / REJECTED Terminal state. HIRED automatically sets JobListing status to FILLED. Time-to-hire and conversion metrics updated on Admin dashboard.	Actor Hiring Manager / System Kafka No — terminal

06 Interview Scoresheet Specification [NEW]

After the Google Meet interview concludes, the Hiring Manager returns to HireFlow and completes the structured scoresheet on the InterviewSlot record. The application cannot advance from INTERVIEW_SCHEDULED until review_submitted_at is set.

Criterion	What is Assessed	Score Range
Technical Skills	Candidate's proficiency in the technical competencies listed in the job description. Includes domain knowledge, tools, and problem-solving ability.	1 – 5

Behavioural	Demonstration of teamwork, adaptability, conflict resolution, and past behaviour in professional scenarios (STAR-method responses).	1 – 5
Communication	Clarity, structure, and confidence in articulating thoughts. Active listening and ability to explain complex ideas.	1 – 5
Culture Fit	Alignment with company values, working style, and team dynamics as assessed by the interviewer.	1 – 5
Problem Solving	Approach to ambiguous or novel challenges. Logical thinking, creativity, and ability to work through unknowns.	1 – 5

Additional Scoresheet Fields

Overall Score	Auto-calculated average of all five criteria scores (displayed to Hiring Manager).
Overall Recommendation	Free-text field — Hiring Manager's narrative summary of the candidate's performance.
Decision	PASS or REJECT — required field. Cannot be left blank.
Rejection Reason	Required if decision is REJECT. Mandatory text input. Stored in ApplicationStageUpdate.reason.
review_submitted_at	Timestamp auto-set by System when scoresheet is submitted. Gates stage advancement.

07 AI Screening — Match Percentage [NEW]

When an application enters SCREENING, the AI model analyses the candidate's resume against the job listing's required skills and qualifications. The result is stored in AIScreeningResult and used in two ways: to inform the Hiring Manager's review, and to generate automated rejection email content.

Field	Description
match_percentage	Integer 0–100. Represents how many of the job's required skills/qualifications the candidate meets. Displayed prominently on the applicant card in the Screening queue.
matched_skills[]	List of skills from the job listing that the candidate's resume satisfies. Shown to Hiring Manager for context.
unmatched_skills[]	List of required skills the candidate is missing. Used in automated rejection email body: 'Your application did not meet the following requirements: [list].'
summary_note	Short AI-generated narrative (2–3 sentences) explaining the score. Visible to Hiring Manager only — never shown to applicant.

screened_at	Timestamp when AI screening completed.
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Screening Decision Logic

Match %	Action	Owner
75% – 100%	Auto-advance to INTERVIEW_SCHEDULED queue	System
40% – 74%	Flagged for Hiring Manager manual review	Hiring Manager
Below 40%	Auto-reject. Notification sent with unmatched_skills[] listed.	System

Note: Thresholds (75%, 40%) are configurable per company by Admin. These are default values.

08 Rejection Flows [NEW/UPDATED]

Automated Rejection (System)

Triggered when AI match_percentage falls below the threshold (default: 40%). The system automatically moves the application to REJECTED and sends a notification email.

Automated Rejection Email — Content

Subject: Your application for [Job Title] at [Company]

Current Stage: REJECTED

Next Stage: N/A — Application Closed

Body includes:

- A thank-you for applying.
- Statement that the application did not meet the role's requirements.
- Specific list of unmatched skills from AIScreeningResult.unmatched_skills[].
- Encouragement to apply for future roles.

Manual Rejection (Hiring Manager)

When a Hiring Manager rejects an application at any stage (Screening review, post-interview), the system presents a mandatory reason input field. The decision cannot be saved without it.

Trigger	Hiring Manager clicks Reject on any application card.
UI Behaviour	System prompts a modal with a required text field: 'Rejection reason (required)'. Save button disabled until field is filled.
Storage	Reason saved to ApplicationStageUpdate.reason. Also stored in AIScreeningResult if rejection happens during Screening.

Notification	Email sent to applicant with current stage (REJECTED), acknowledgement that a human reviewed their application, and the reason provided. Next stage: N/A.
Audit	Actor ID, timestamp, and reason permanently logged. Cannot be edited post-submission.

09 Notification Email Specification [UPDATED]

Every notification triggered by a Kafka `application.stage.changed` event must include the applicant's current stage AND their next expected stage. This applies to all stage transitions without exception.

Transition	Email Subject / Content	Stage Block in Email
APPLIED → SCREENING	Application received & under review	Current: APPLIED Next: SCREENING (AI review in progress)
SCREENING → INTERVIEW SCHEDULED	Congratulations — interview invite + Google Meet link + slot details	Current: INTERVIEW SCHEDULED Next: OFFER SENT (if successful)
SCREENING → REJECTED (Auto)	Application unsuccessful + specific unmatched skills listed	Current: REJECTED Next: N/A
SCREENING → REJECTED (Manual)	Application unsuccessful after human review + reason provided	Current: REJECTED Next: N/A
INTERVIEW → OFFER SENT	Offer details incoming	Current: OFFER SENT Next: HIRED (pending acceptance)
INTERVIEW → REJECTED	Post-interview outcome + mandatory reason from HM	Current: REJECTED Next: N/A
OFFER SENT → HIRED	Welcome to the team	Current: HIRED Next: Onboarding (external)

10 Problem Statements & Proposed Solutions

Each problem maps directly to proposed solutions, the affected actor, and the system owner.

P1 Slow Hiring Cycle

Affected Actor: Hiring Manager / Admin

Manual reviews create bottlenecks. Admins have no visibility into where slowdowns occur.

Proposed Solution

Owner: System (AI) + Hiring Manager + Admin

- AI scoring auto-advances clear matches, reducing manual review load.
- Hiring Manager receives pre-ranked, pre-filtered list with match percentages.
- Admin dashboard surfaces time-to-hire per stage in real time.
- Reminder nudges sent to Hiring Manager for pending reviews.

Data Model: `ApplicationStageUpdate.created_at` enables per-stage duration calculation.

P2 Poor Candidate Communication

[UPDATED]

Affected Actor: Applicant

Applicants submit and hear nothing. No stage updates, no reasons, no timeline expectations.

Proposed Solution

Owner: System (Kafka)

- Kafka event on every stage change → immediate notification to applicant.
- Every email includes current stage AND next expected stage.
- Auto-rejections include specific unmatched skills. Manual rejections include HM reason.
- Interview invite includes Meet link + slot details automatically.

Data Model: `ApplicationStageUpdate.next_stage` added to event payload.

P3 No Visibility into Job Listing State

[NEW]

Affected Actor: Admin / Hiring Manager

No way to know the current state of a job posting — whether it's live, paused, filled, or draft.

Proposed Solution

Owner: System + Admin

- Job Listing now has a status field: DRAFT, OPEN, PAUSED, CLOSED, FILLED.
- Status visible on Admin and Hiring Manager dashboards at all times.
- FILLED status auto-set by System when a candidate reaches HIRED.
- Admin can manually transition listings between OPEN, PAUSED, and CLOSED.

Data Model: `JobListing.status (Enum)`: DRAFT | OPEN | PAUSED | CLOSED | FILLED.

P4 Unstructured Post-Interview Feedback

[NEW]

Affected Actor: Hiring Manager

Reviewers return from Google Meet with no standard format, making candidate comparison impossible.

Proposed Solution

Owner: Hiring Manager

- Structured scoresheet with 5 criteria: Technical, Behavioural, Communication, Culture Fit, Problem Solving.
- Each criterion scored 1–5. Overall score auto-calculated.
- Mandatory Overall Recommendation (free text) and Decision (PASS/REJECT).
- Rejection requires mandatory reason input before scoresheet can be saved.
- Application gates on review_submitted_at — cannot advance without submitted scoresheet.

Data Model: InterviewSlot extended with all scoresheet fields + review_submitted_at.

P5 No AI Transparency During Screening

[NEW]

Affected Actor: Hiring Manager

Hiring Managers review applications with no indication of how well a candidate matches the role.

Proposed Solution

Owner: System (AI) + Hiring Manager

- AI match percentage (0–100%) displayed on each applicant card in Screening queue.
- Matched and unmatched skills shown for context.
- AI summary note (2–3 sentences) visible to Hiring Manager only.
- Auto-reject threshold (default 40%) configurable by Admin per company.

Data Model: AIScreeningResult entity: match_percentage, matched_skills[], unmatched_skills[], summary_note.

P6 Inconsistent Rejection Communication

[NEW]

Affected Actor: Applicant / Hiring Manager

Automated rejections give no detail. Manual rejections have no enforced standard.

Proposed Solution

Owner: System + Hiring Manager

- Automated rejection email lists specific unmatched_skills[] from AIScreeningResult.
- Manual rejection triggers mandatory reason field — cannot be skipped.
- Both rejection types send notification with current + next stage and reason surfaced.
- All rejection reasons permanently stored in ApplicationStageUpdate for audit.

Data Model: ApplicationStageUpdate.reason (required on rejection). AIScreeningResult.unmatched_skills[] in email.

11 Functional Requirements by Actor

Applicant

- Register and create a profile with personal details and resume upload.
- Browse OPEN job listings and apply to open roles.
- System prevents applying to the same role twice — clear error returned.
- Receive real-time notifications on every stage change. Every email includes current stage and next expected stage.
- Auto-rejection emails list specific skills that did not match the job requirements.
- View current application status and history on applicant dashboard.

Hiring Manager

- View all applications in Screening queue with AI match percentage, matched skills, and summary note.
- Approve or reject screened applications. Manual rejections require a mandatory reason (cannot be skipped).
- Rescan / re-review individual resumes at any point during screening.
- Create InterviewSlots: date, time, duration, and Google Meet link (required field).
- System auto-sends slot details and Meet link to applicant on slot creation.
- Submit structured post-interview scoresheet: 5 criteria scored 1–5, Overall Recommendation, Decision.
- Decision of REJECT on scoresheet triggers mandatory rejection reason prompt.
- Application cannot advance from INTERVIEW_SCHEDULED without a submitted scoresheet.
- View and manage JobListing status (DRAFT, OPEN, PAUSED, CLOSED).

Admin

- Dashboard: all active listings with current JobListing.status and applicant counts.
- Pipeline funnel: application volume per stage across all listings.
- Time-to-hire metrics filterable by role and date range.
- Configure AI match threshold (default 40%) per company.
- Access full ApplicationStageUpdate audit log including all rejection reasons.
- Manually transition JobListing status between OPEN, PAUSED, and CLOSED.

System

- On APPLIED: parse resume, enforce duplicate constraint, publish Kafka event.
- On SCREENING: run AI model, calculate match_percentage, populate AIScreeningResult.
- Auto-advance applications above threshold; auto-reject below threshold with unmatched skills in notification.
- Every Kafka event payload includes: current_stage, next_stage, reason (if rejection).
- On HIRED: automatically set JobListing.status = FILLED.
- Enforce UNIQUE index on Application(applicant_id, job_listing_id).
- Gate stage advancement from INTERVIEW_SCHEDULED on InterviewSlot.review_submitted_at.

- Prevent backwards stage transitions at API layer.

12 Non-Functional Requirements

Event Reliability	Kafka at-least-once delivery. Consumers idempotent on event_id.
Notification Completeness	Every notification email must include current_stage and next_stage fields. No exceptions.
Duplicate Guard	Unique constraint at DB index + API validation layers.
Rejection Reason Gate	API rejects REJECT decision payload if reason field is null or empty string.
Scoresheet Gate	Stage transition from INTERVIEW_SCHEDULED blocked if review_submitted_at is null.
Audit Trail	ApplicationStageUpdate is append-only. Reason field immutable post-submission.
AI Transparency	match_percentage and summary_note visible to Hiring Manager only. Unmatched skills visible in rejection emails to applicant.
Stage Ordering	Backwards transitions rejected at API layer. Forward-only state machine.
Dashboard Freshness	Admin metrics lag no more than 30 seconds behind latest ApplicationStageUpdate.
Multi-tenancy	All queries scoped by company_id. AI thresholds configurable per company.
Meet Link Validation	meet_link validated as URL format. InterviewSlot cannot be saved without it.

13 Kafka Event Specification [UPDATED]

Topic: application.stage.changed — published on every Application stage transition.

Field	Type	Description
event_id	UUID	Idempotency key for consumers
application_id	UUID	FK → Application
applicant_id	UUID	FK → User (applicant)
job_listing_id	UUID	FK → JobListing
company_id	UUID	Tenant scoping

previous_stage	Enum	Stage before transition
current_stage	Enum	Stage after transition
next_stage	Enum?	[NEW] Next expected stage — included in all notification emails. Null on terminal stages.
actor_id	UUID	User who triggered the transition
actor_role	Enum	APPLICANT HIRING_MANAGER SYSTEM
reason	String?	Rejection reason (required when new_stage = REJECTED)
unmatched_skills	String[]?	[NEW] From AIScreeningResult — included on auto-rejection events
occurred_at	ISO 8601	UTC timestamp of transition

14 Out of Scope — MVP

- x Google Calendar / Meet API integration (auto-generate Meet links) — Phase 2.
- x Candidate offer acceptance workflow inside HireFlow — external process at MVP.
- x Background check integration between OFFER_SENT and HIRED.
- x Referral tracking and sourcing attribution.
- x HRIS / payroll sync on HIRED status.
- x Bias auditing on AI screener — flagged for Phase 2 compliance review.
- x Multi-interviewer panel support — single reviewer per InterviewSlot at MVP.
- x Scoresheet template customisation per role — single default template at MVP.

15 Open Questions & Assumptions

Topic	Question / Assumption
AI Model	Internally trained or third-party API? Latency SLA for match_percentage to be defined.
Score Thresholds	Default 40% auto-reject and 75% auto-pass — need product + stakeholder sign-off before build.
Scoresheet Criteria	Are the 5 criteria fixed platform-wide, or configurable per company / role at MVP?
Review Deadline	How long before unsubmitted scoresheet triggers escalation? Suggested: 48 hrs.
Offer Acceptance	Applicant accepts/declines offer inside HireFlow or via external email? MVP assumption: external.

Role Expiry

Does JobListing auto-close after a deadline? Not yet specified — needs decision.

Data Retention

Applicant data deletion / GDPR / NDPR compliance not yet scoped. Pre-production blocker.